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Prevalence and factors associated with informal advance care planning discussion and do-not-resuscitate directives in patients at geriatric clinics

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ABSTRACT

Background: Advance care planning (ACP) discussions and do-not-resuscitate (DNR) directives are essential for ensuring quality end-of-life care, especially for older adults with multiple comorbidities. This study aimed to investigate the prevalence and associated factors of informal ACP discussions and DNR directives among geriatric outpatients in Taiwan.

Methods: A cross-sectional study was conducted among 276 Taiwanese geriatric outpatients aged 65 years and older. Data on demographics, comorbidities, functional status, informal ACP discussions, and DNR directives were collected. Multinomial logistic regression was used to analyze the association between independent variables and dependent variables (ACP discussion experience and DNR directives).

Results: While 92.1 % of participants reported having informal ACP discussions, only 8.3 % had completed DNR directives. Compared to those who only discussed ACP (reference group), individuals with a one-point higher Cumulative Illness Rating Scale for Geriatrics score were significantly more likely to have both discussed ACP and completed DNR directives (odds ratio [OR] = 1.14, 95 % confidence interval [CI]: 1.01–1.29, $p = .035$). Conversely, individuals with junior high school education or higher (OR = 0.25, 95 % CI: 0.08–0.84, $p = .025$) and those dependent on others for activities of daily living (ADL dependence; OR = 0.24, 95 % CI: 0.07–0.86, $p = .029$) were significantly less likely to have neither informally discussed ACP nor completed DNR directives.

Conclusion: While Taiwanese older outpatients often informally discussed ACP, the completion of DNR directives was less common. The link between higher geriatric comorbidity and DNR directives highlights the need for proactive, tailored interventions in this population.

1. Introduction

As older adults are living longer and healthier lives, the need for high-quality end-of-life (EOL) care in healthcare systems is being increasingly recognized [1]. Aging is often accompanied by multimorbidity (i.e., having two or more chronic conditions), frailty, and cognitive decline, and all can impair individuals' ability to make informed medical decisions about EOL care [2]. While these circumstances necessitate critical treatment-related decisions, many older

adults struggle to actively participate in these discussions, highlighting the importance of early engagement in EOL care planning before cognitive and physical decline limits the ability to express preferences [3].

Advance care planning (ACP) enables individuals to define their future medical care goals, discuss them with family and healthcare providers, and document their wishes in formal directives [4]. Schickel et al. [5] conceptualized ACP as a staged process: (1) contemplating values and treatment preferences, (2) discussing them with

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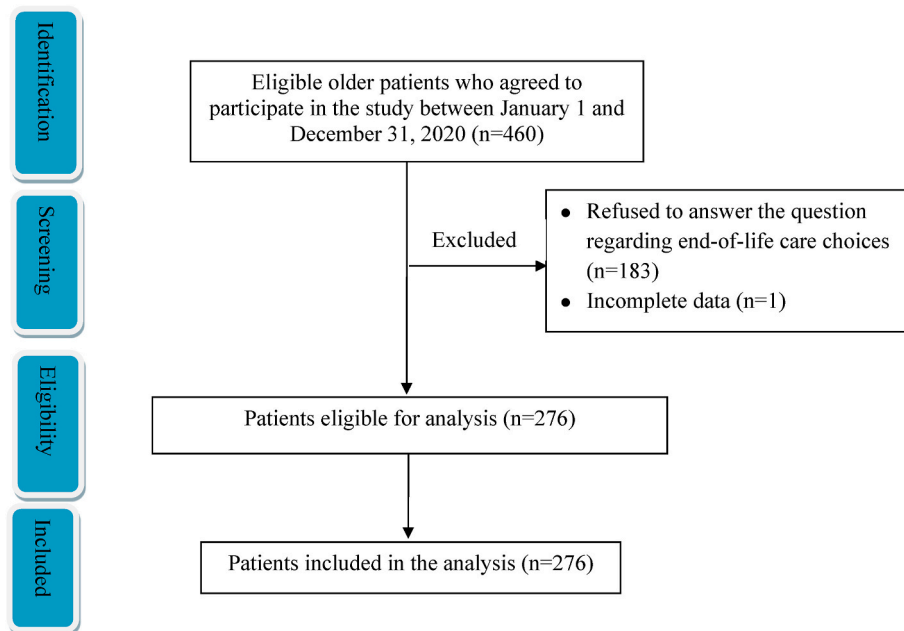


Fig. 1. STROBE flow diagram for participant recruitment.

family and friends, (3) engaging in conversations with healthcare providers, and (4) completing formal advance directives (ADs; e.g., living wills or durable powers of attorney). Given that older adults may be at different stages of this process, personalized support is essential to facilitate the transition from initial discussions to formalized ACP decisions.

Despite the benefits, ACP engagement and AD completion remain challenging, particularly for older adults with comorbidity. Compared to younger generations, older adults are more concerned about the burdens ACP discussions may place on their families, including potential conflicts and emotional distress [6]. Additionally, cultural norms in Confucian-influenced societies often discourage direct discussions about EOL care [7]. While the prevalence of ACP engagement in Western countries ranges from 65 to 81 % and AD completion is reportedly 36.7 % in the United States of America [8–10], these figures are notably lower in Asian countries [11–13]. For example, in Japan, while 72 % of the general public supports ACP discussions, only 8.7 % have completed an AD [13,14]. In response, Moris et al. [7] proposed a consensus definition of ACP and provided recommendations for patient-centered and family-based initiatives in Asia, as follows:

[...] a process that enables individuals to identify their values, to define goals and preferences for future medical treatment and care, to discuss these values, goals, and preferences with family and/or other closely related persons, and health-care providers, and to record and review these preferences if appropriate.

In Taiwan, legal and policy efforts have been implemented to enhance patient autonomy and EOL care quality. The Hospice Palliative Care Act (Natural Death Act) was enacted in 2000, granting patients the right to refuse resuscitation and establish do-not-resuscitate (DNR) directives [15]. The Patient Right to Autonomy Act, legalized in 2016 and enacted in 2019, mandates that formal ACP counseling must involve a multidisciplinary team (physician, nurse, social worker, and clinical psychologist) and a structured consultation process [16]. These legal frameworks aim to promote ACP and increase patient-provider discussions about EOL preferences. However, it remains unclear whether these laws have facilitated informal ACP discussions with family, friends, and healthcare providers.

Informal ACP discussions, defined as conversations about future medical care preferences without legal documentation, play a crucial

role in shaping individuals' readiness for formal ACP and DNR directives [5]. Prior studies suggest that various demographic, psychosocial, and functional factors influence ACP engagement [3,6,17–20], yet most research relies on self-reported data, raising concerns about measurement accuracy [8,9,11,21]. Additionally, while both informal ACP discussions and DNR directives contribute to increased confidence in decision-making, the factors influencing each process may differ, particularly in cultural contexts like Taiwan [9]. Notwithstanding, little research has explored the relationship between comorbidity severity, informal ACP discussion experience, and DNR directives.

Although ACP legislation primarily focuses on formal ACP discussion and documentation, ACP is widely considered as a dynamic communication process rather than a singular decision-making event [4,5]. From a clinical and behavioral perspective, informal ACP discussions often precede formal ACP engagement, helping individuals clarify their values and preferences before seeking formal counseling. Given the low uptake of formal ACP counseling in Taiwan [22], we considered that focusing solely on formal ACP would limit our ability to assess ACP engagement patterns. Therefore, this study adopted a broader approach by examining informal ACP discussions as a key component of the ACP process.

To address the aforementioned knowledge gaps, this study aimed to investigate the prevalence of informal ACP discussions and DNR directives among older adults in Taiwan. We also examined the functional factors associated with informal ACP and DNR engagement using data from the Comprehensive Geriatric Assessment, which includes validated instruments to enhance measurement accuracy and minimize information bias. Understanding the prevalence and determinants of informal ACP discussions can provide insights into strategies that promote the transition from informal to formal ACP participation.

2. Methods

2.1. Study design

This cross-sectional study was reported according to the STROBE guidelines (Appendix 1) [23].

Ethical Approval

The Institutional Review Board of National Cheng Kung University

Table 1

Demographic characteristics, comorbidities, and functional status of participants.

Variables	Total (N = 276)	Neither informal ACP nor DNR directives (n = 22)	Informal ACP only (n = 231)	Both informal ACP and DNR directives (n = 23)	p value
Age, mean(SD)	79.89 (6.95)	79.50 (8.35)	79.92 (6.83)	79.96 (7.00)	F = 0.04, p = 0.963
Male, n(%)	105 (38)	8 (36.4)	90 (39)	7 (30.4)	$\chi^2 =$ 0.67, p = 0.714
Educational level of junior high school and higher, n (%)	116 (42)	4 (18.2)	103 (44.6)	9 (39.1)	$\chi^2 =$ 5.84, p = 0.054
CIRS-G score, mean(SD)	6.74 (3.84)	7.59 (3.29)	6.53 (3.92)	8.09 (3.20)	F = 2.33, p = 0.099
Frailty, n(%)	57 (20.7)	4 (18.2)	47 (20.4)	6 (26.1)	$\chi^2 =$ 0.51, p = 0.783
ADL dependence, n(%)	131 (47.5)	6 (27.3)	112 (48.5)	13 (56.5)	$\chi^2 =$ 4.45, p = 0.108
IADL dependence, n(%)	158 (57.3)	16 (72.7)	129 (55.8)	13 (56.5)	$\chi^2 =$ 2.35, p = 0.310
Risk of malnutrition and malnutrition, n(%)	126 (45.7)	13 (59.1)	101 (43.7)	12 (52.2)	$\chi^2 =$ 2.34, p = 0.310
Cognitive function abnormality, n(%)	70 (25.4)	9 (40.9)	54 (23.4)	7 (30.4)	$\chi^2 =$ 3.54, p = 0.170
Depression, n (%)	69 (25)	4 (18.2)	60 (26)	5 (21.7)	$\chi^2 =$ 0.79, p = 0.673

ACP, advance care planning; ADL, activities of daily living; CIRS-G, Cumulative Illness Rating Scale for Geriatrics; DNR, do-not-resuscitate; IADL, instrumental activities of daily living.

Hospital approved this study. All participants signed informed consent forms before data collection.

2.2. Setting and participants

Participants were recruited from the geriatric outpatient clinic of a 1193-bed tertiary care university hospital in Tainan, Taiwan, from January 1 to December 31, 2020. Inclusion criteria were being aged ≥ 65 years, having visited the geriatric outpatient clinic for the first time (i.e., since we routinely evaluated first-time outpatients using the Comprehensive Geriatric Assessment, allowing us to obtain baseline patient functional data before any geriatric intervention), and being able to communicate with nursing managers. Exclusion criteria were having a severe cognitive/mental disorder, an inability to communicate clearly, and not responding to the interview despite providing informed consent.

2.3. Dependent variables

The dependent variables were informal ACP discussion experience and DNR directives. To differentiate informal ACP discussion experience from formal ACP counseling (i.e., which is regulated by the Patient Right to Autonomy Act), we defined informal ACP discussions as conversations conducted with significant others and/or healthcare providers about future medical care in case of serious illnesses and that do not involve formal documentation or structured medical guidance. Informal ACP discussion experience was measured using a single question: “*In the past year, have you ever discussed with anyone the care or medical treatment you would want to receive if you became seriously ill in the future?*” Response options were “yes,” “no,” “don’t know,” and “refuse to answer.” The latter three options were grouped into a single category, “otherwise.” Given the potential sensitivity of discussions pertaining to EOL care among Taiwanese older adults, participants were first asked if they were willing to answer a question regarding EOL care choices, and only those who consented were asked about their informal ACP discussion experience.

DNR directives were measured by checking for DNR status in the hospital’s electronic medical record system. The DNR status results were dichotomized into “yes” or “no.” According to the regulations of the National Health Insurance Administration in Taiwan, after a patient completes a legitimate DNR directive, one’s DNR choices are registered on the patient’s health insurance IC card, enabling all national health-care facilities to identify these choices and provide care accordingly. At the study hospital, we registered the patients’ DNR directives in the electronic medical record system by reading their health insurance IC cards upon their hospital visit.

We divided the participants into three groups: (1) neither informal ACP discussion nor DNR directives, (2) informal ACP discussion without DNR directives (hereinafter informal ACP only), and (3) informal ACP discussion and DNR directives.

2.4. Independent variables and measurement

Demographic data were collected upon participant inclusion. Age was a continuous variable, sex a dichotomous variable (male and female), and educational level was dichotomized as higher or lower than junior high school.

Clinical data were also collected. To assess the overall frailty level after participants had been evaluated by a healthcare professional, we used the nine-point Clinical Frailty Scale [24], where higher scores indicate greater frailty risk. Moreover, to simplify the measurement of comorbid illnesses and their severities under a single numerical score and enable comparisons between participants, we used the Cumulative Illness Rating Scale for Geriatrics (CIRS-G; higher scores indicate a greater burden of comorbidities) [25], which is a good predictor of mortality and morbidity in older patients.

Functional data were also collected. ADL was assessed using the Katz ADL, an ordinal scale used to measure ADL performance [26]; six variables describing ADLs (eating, bathing, dressing, toileting, transferring, and continence) were scored with either 0 (dependence) or 1 (independence) and ADL function was dichotomized into independence or dependence (if any item was responded with dependence). To measure IADL, we used the Lawton IADL Scale and assessed the following abilities: using the telephone, shopping, food preparation, housekeeping, laundry, transportation, responsibility for own medications, and handling finances [27]. We dichotomized IADL function into independence or dependence (if any item was responded with dependence). Cognitive function was measured using the 10-item Short Portable Mental Status Questionnaire, a tool designed to screen for cognitive impairment in geriatric inpatients/outpatients [28]; cognitive function was dichotomized into normal and abnormal (having more than three errors in 10 questions). Mood was measured by the five-item Geriatric Depression Scale and dichotomized into normal and depression (if score

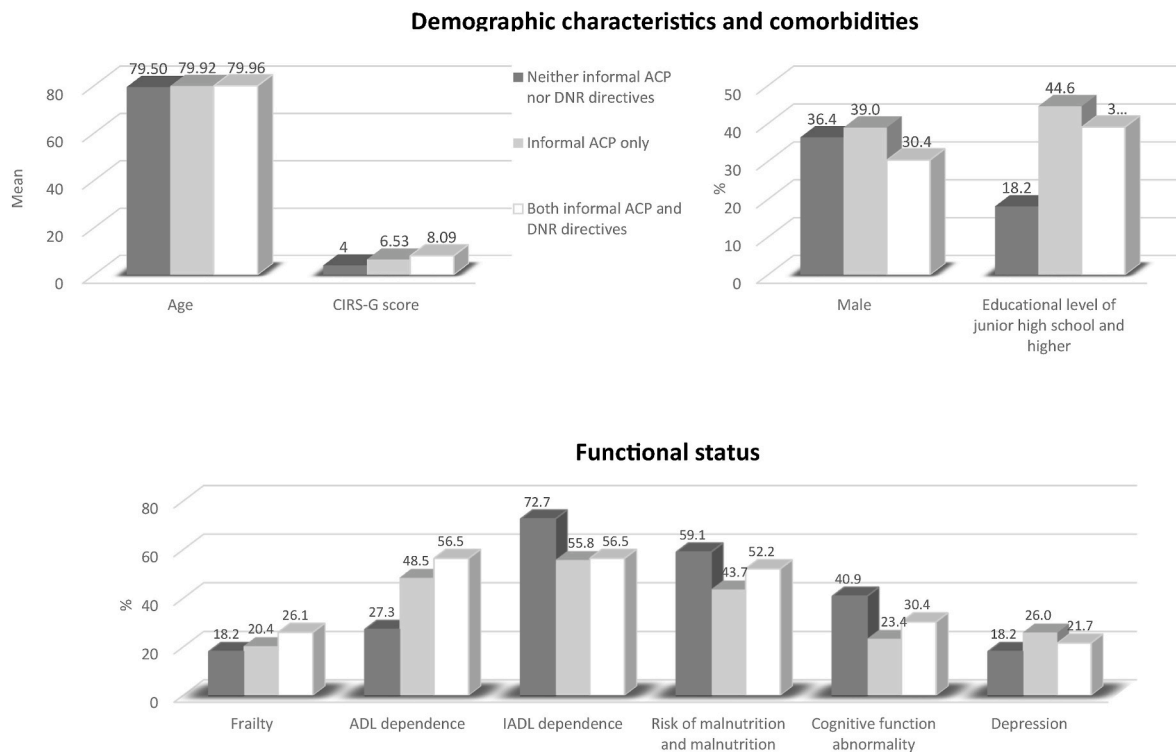


Fig. 2. Distribution of participants' demographic characteristics, comorbidities, and functional impairment.

≥ 2) [29]. Nutritional status was measured according to the Mini Nutritional Assessment–Short Form, a screening tool for identifying patients who are malnourished/at risk of malnutrition [30], and was dichotomized into normal versus malnutrition and risk of malnutrition (if score ≤ 7 ; maximum score of 14 points).

All independent variable data were collected through face-to-face assessments conducted by three geriatric nurses with at least two years of experience in performing the Comprehensive Geriatric Assessment.

2.5. Statistical methods

Continuous variables were presented as means with standard deviations, and categorical variables as counts with percentages. Comparisons of independent variables by dependent variable groups were performed using χ^2 test or Fisher's exact test.

A multinomial logistic regression model was used for multivariable analyses; all independent variables were entered simultaneously to adjust for each other and control for confounders (age, sex, and educational level). The reference group was the informal ACP only group, which produced adjusted odds ratios (ORs) for each independent variable of interest and its association with informal ACP discussion and DNR document status. Variables were checked for multicollinearity, and all variance inflation factors were acceptable (< 2.0). Adjusted ORs and 95 % confidence intervals (CI) were calculated for the multivariate-adjusted models, and statistical significance was represented by $p < 0.05$. Model fit was assessed using design-based goodness-of-fit for survey logistic regression. All analyses were performed using SPSS version 22.0 (SPSS Inc.).

3. Results

3.1. Participants' characteristics

Initially, 460 patients were included. Those who refused to answer the question regarding EOL care choices ($n = 183$) or had incomplete

data ($n = 1$) were excluded; the data of 276 patients were analyzed (Fig. 1). Compared to excluded patients, included patients had a significantly higher prevalence of frailty (20.7 % vs. 10.9 %, $p = .006$) and ADL dependence (47.5 % vs. 35.5 %, $p = .011$). No significant differences were observed in age, gender, educational level, CIRS-G score, IADL dependence, malnutrition risk, cognitive function abnormality, or depression (Appendix 2).

Participants had a mean age of 79.89 ± 6.95 years (range, 65–98 years), most were women (62 %), and 68 (24.6 %) had no formal education. The mean CIRS-G and Clinical Frailty Scale score was 6.74 ± 3.84 (range, 0–20) and 4.48 ± 1.13 (range, 3–6), respectively. Regarding functional status, 47.5 % ($n = 131$) and 57.3 % ($n = 158$) of participants had ADL dependence and/or IADL dependence, respectively. The malnutrition and risk of malnutrition rate was 45.7 %, and the average Mini Nutritional Assessment–Short Form score was 11.43 ± 2.29 (range, 3–14). Cognitive impairment and depression rates were at 25.4 % ($n = 70$) and 25 % ($n = 69$), respectively (Table 1). Fig. 2 shows the distribution of participants' demographic characteristics, comorbidities, and functional impairment.

3.2. Prevalence of informal ACP discussion and DNR directives

In the past year, 92.1 % ($n = 254$) of the participants had informal ACP discussion experience, and the prevalence of DNR directives among them was 8.3 % ($n = 23$). All participants with DNR directives had informal ACP discussion experience. Of the 23 with DNR directives, 65.2 % ($n = 15$) chose to forgo life-sustaining treatments and 82.6 % ($n = 19$) agreed to receive hospice EOL care.

3.3. Factors associated with informal ACP discussion and DNR directives

The group with neither informal ACP discussion nor DNR directives had fewer participants with higher than junior high school education (18.2 %) than did the groups with ACP only (44.6 %) and ACP discussion and DNR directives (39.1 %). Regarding ADL dependence, the rate was 27.3 %, 48.5 %, and 56.5 % in the groups with neither informal ACP

Table 2

Multinomial logistic regression predicting informal advance care planning discussion experience, do-not-resuscitate directives, and both advance care planning discussion experience and do-not-resuscitate directives.

Variables	Neither informal ACP nor DNR directives			Both informal ACP and DNR directives		
	B	Odds ratio (95 % CI)	p	B	Odds ratio (95 % CI)	p
Age	−0.04	0.96 (0.89, 1.04)	0.354	−0.02	0.98 (0.92, 1.06)	0.646
Male ^a	−0.24	0.79 (0.27, 2.30)	0.660	−0.46	0.63 (0.22, 1.79)	0.386
Educational level of junior high school or higher ^b	−1.38	0.25 (0.08, 0.84)	0.025	−0.02	0.98 (0.37, 2.61)	0.977
CIRS-G score	0.10	1.11 (0.97, 1.26)	0.135	0.13	1.14 (1.01, 1.29)	0.035
Frailty ^c	−0.64	0.53 (0.10, 2.72)	0.445	−0.13	0.88 (0.22, 3.56)	0.855
ADL dependence ^c	−1.42	0.24 (0.07, 0.86)	0.029	0.19	1.21 (0.43, 3.41)	0.715
IADL dependence ^c	0.89	2.43 (0.69, 8.61)	0.168	−0.38	0.68 (0.21, 2.21)	0.542
Malnutrition and risk of malnutrition ^c	0.82	2.26 (0.76, 6.71)	0.141	0.53	1.69 (0.61, 4.73)	0.316
Cognitive function abnormality ^c	0.52	1.69 (0.54, 5.30)	0.369	0.23	1.26 (0.40, 3.99)	0.695
Depression ^c	−0.63	0.53 (0.16, 1.80)	0.313	−0.24	0.78 (0.26, 2.35)	0.663

ACP, advance care planning; ADL, activities of daily living; CIRS-G, Cumulative Illness Rating Scale for Geriatrics; DNR, do-not-resuscitate; IADL, instrumental activities of daily living.

Referral group: informal ACP only.

^a Referral group: female.

^b referral group: elementary school education or less.

^c Referral group: without functional impairments.

discussion nor DNR directives, ACP only, and ACP discussion and DNR directives, respectively. Univariate analysis showed non-significant differences between the three groups regarding higher than junior high school education ($p = 0.05$) and ADL dependence ($p = 0.11$; Table 1).

The multinomial logistic regression showed significantly reduced ORs (vs. informal ACP only) of having neither informal ACP discussion experience nor DNR directives for participants with a junior high school education and higher (OR = 0.25, 95 %CI = 0.08–0.84, $p = 0.025$) and ADL dependence (OR = 0.24, 95 %CI = 0.07–0.86, $p = 0.029$). Moreover, a one-point increase in the CIRS-G score was associated with significantly increased ORs of having both ACP discussion experience and DNR directives (OR = 1.14, 95 %CI = 1.01–1.29, $p = 0.035$). Age, sex, frailty status, IADL dependence, malnutrition and risk of malnutrition, cognitive function abnormality, and depression were not significantly different among the three groups (Table 2). A forest plot showing the odds ratio of independent variables in the multinomial regression model is shown in Fig. 3.

4. Discussion

Our study describes the prevalence of ACP discussion experience and

DNR directives at geriatric outpatient clinics in Taiwan and their association with demographic, comorbidity, and functional factors. Overall, 92.1 % of participants reported informal ACP discussion experience, while only 8.3 % had DNR directives. Moreover, compared with participants with informal ACP only, those with a higher educational level and ADL dependence were associated with a lower likelihood of having neither informal ACP discussion experience nor DNR directives. Meanwhile, higher CIRS-G scores were associated with a higher likelihood of DNR directives. This is the first Asian study to simultaneously investigate the factors associated with informal ACP discussion experience and DNR directives using the same sample. As there are cultural variations between countries in the acknowledgment of/desire for ACP and DNR directives [16,19], our results help inform healthcare providers in delivering culturally-sensitive care for older adults, especially in Taiwan.

Although informal ACP discussions may not always lead to formal medical decisions, evidence shows them as crucial in shaping individuals' readiness for formal ACP and DNR directives [4,5]. According to Schickedanz et al.'s ACP framework,⁵ ACP is a process that starts with individuals contemplating their future medical care values/preferences, involves discussions of these preferences with family/friends and healthcare providers, and ends in the creation of formal documents (e.g., DNR directives). That is, informal ACP discussions occur in the early stages of the ACP process, laying the groundwork for formal ACP engagement and facilitating the completion of DNR directives; such informal processes allow for preference and value discussions, concerns about ACP to be addressed, the identification of potential surrogate decision-makers [5].

The prevalence of informal ACP discussion experience in our study (92.1 %) was higher than that in past American and Japanese studies [8, 9,13]. For example, it was at 65 % among American adults aged 55–74 years [9], 81 % among those aged >50 years in the Health and Retirement Study [8], and 72 % among Japanese participants aged 40–75 years [13]. Regarding possible explanations for our findings, previous studies have shown that it is more feasible for Chinese older adults with chronic diseases to engage in ACP discussions [3]. Moreover, legislation and policy in Taiwan may have promoted ACP discussion. After the legislation in 2016 and the enactment of the Patient Right to Autonomy Act in 2019, media and stakeholder groups generated considerable propaganda. Still, in Taiwan, out-of-pocket payments are mandatory for formal ACP consultations, potentially barring the access of certain populations (e.g., those with lower socioeconomic status) to such consultations and leading to a preference toward informal (vs. formal) ACP discussions with family/friends/healthcare providers. This contextualization renders relatively unsurprising the finding of a high prevalence of informal ACP discussion experience in our sample. Therefore, financial limitations may encourage engagement in informal ACP discussions but concomitantly hinder older adults' access to comprehensive ACP services, ability to make fully informed EOL decisions, and to legally document their EOL care preferences.

Despite the high prevalence of ACP discussion experience, only 8.3 % of the participants had DNR directives. In the United States of America [10] and Australia [31], the population-based prevalence of AD documentation was 36.7 % and 14 %, respectively; in Japan [14] and South Korea [11], it was 8.7 % and 4.7 %, respectively. The low prevalence of DNR directives in our study may be attributed to the strong cultural emphasis on family-based decision-making in Chinese societies. This is partly influenced by the concept of "filial piety," which underscores children's responsibility to care for their parents and make decisions in their best interest. As a result, patient autonomy often takes a secondary role to family influence, making independent decision-making regarding EOL care less common [32]. Balancing family involvement and patient autonomy hence remains a critical challenge.

Additionally, many older Chinese individuals prefer a passive role in decision-making, do not fully recognize the value of active participation in healthcare decisions, and may fear making incorrect choices [19,33].

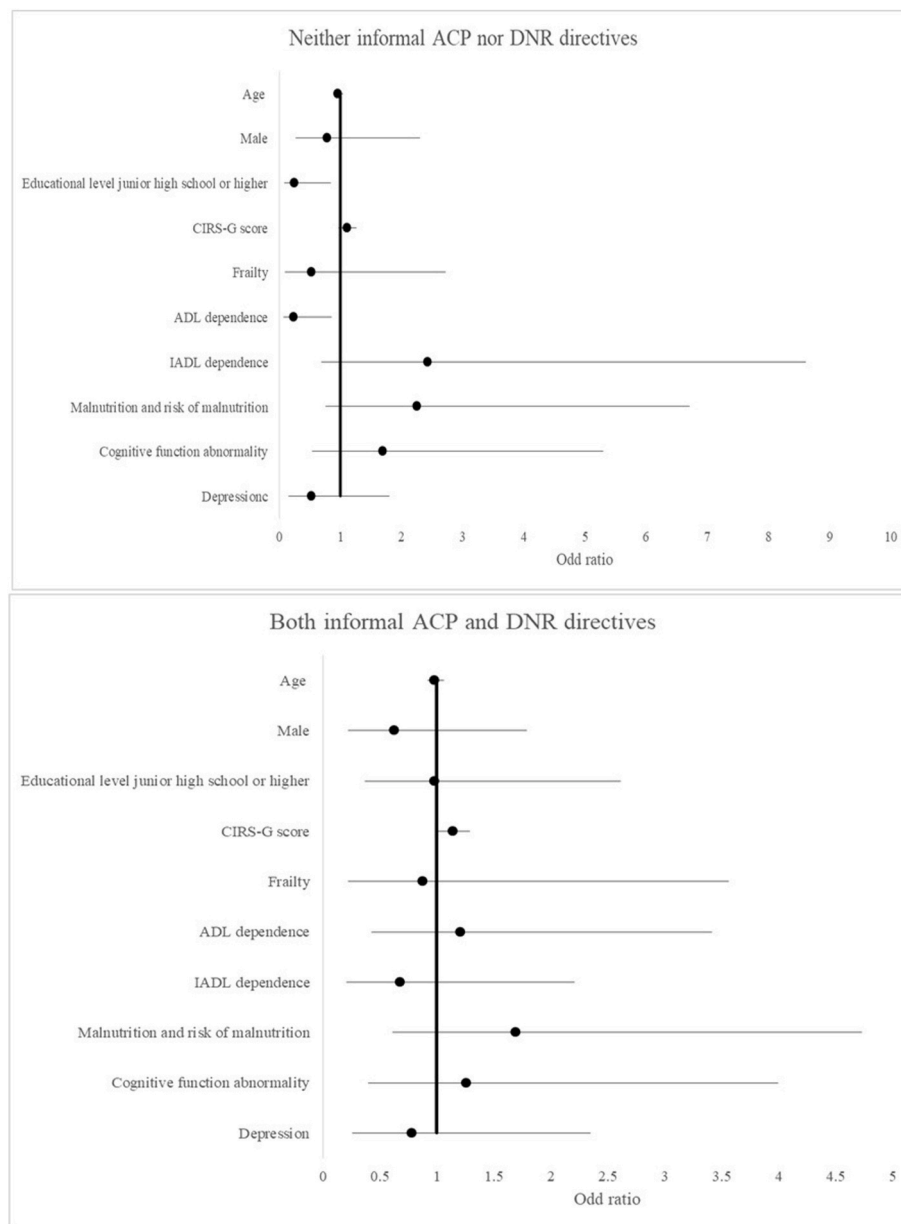


Fig. 3. Forest plot showing the odds ratios of independent variables in the multinomial regression model. (A) Odds ratios of independent variables comparing neither informal ACP discussion experience nor DNR directives to informal ACP only; (B) Odds ratios of independent variables comparing both informal ACP discussion experience and DNR directives to informal ACP only. Note. ACP, advance care planning; DNR, do-not-resuscitate.

In a qualitative study conducted in Hong Kong, many older adults relinquished their right to plan for EOL care and intended to delegate the responsibility to others, hindering their acquisition of DNR directives [34]. Furthermore, when Chinese older adults are asked to designate a surrogate decision-maker, they often choose their physicians or family members, opting out of making decisions independently [34]. Government policies in Taiwan reinforce this tendency by requiring two adult witnesses during DNR directive formalization, which may unintentionally encourage individuals to engage in informal discussions with family members rather than to independently pursue formal DNR directives. Consequently, cultural norms, decision-making preferences, and local policy regulations may have collectively contributed to the low prevalence of DNR directives in this study.

Much evidence shows the positive effect of having a physician who has broached ACP about DNR directives [9,21]. Nevertheless, a recent systematic review highlighted that while healthcare providers in Asia recognize the importance of ACP, they rarely directly get patients

involved and often face difficulties initiating related discussions [35]. These challenges stem from limited knowledge and skills, personal discomfort, fear of family conflict or legal consequences, and the lack of standardized systems. Considering that our participants were first-time visitors to the relevant clinic and the prevalence of DNR directives was low, Chinese geriatricians could further discuss EOL treatment preferences with patients and their family members and encourage formal ACP discussion, DNR directives, and AD registration during subsequent outpatient visitations.

In our study, participants with a higher than junior high school education (vs. with ACP only) were less likely to have neither informal ACP discussion experience nor DNR directives. These findings confirm past evidence, wherein higher education level and ADL dependence were associated with informal ACP discussions [3,17,36–38]. This may be because older adults with a higher education level have higher health literacy, enabling them to better understand their disease prognosis and express treatment preferences [8]. Additionally, participants with ADL

dependence were less likely to have neither informal ACP discussion experience nor DNR directives than participants with ACP only, which is partially supported by the existing literature. Older adults who perceive that their health limits their daily functioning (vs. who do not report functional limitations) have double the likelihood of engaging in ACP discussions [39]. Concomitantly, the evidence on the associations between physical functioning and ACP are mixed [17].

Our study demonstrates that comorbidity severity is associated with DNR directives in participants with previous informal ACP discussion experience. Older adults with chronic diseases (vs. without) are more likely to have DNR directives or AD [11,31,39]. Furthermore, older patients with comorbidity had frequent healthcare utilization and prolonged illness duration, increasing the chance of prior discussion of EOL care preferences with healthcare providers [31,39]. Older patients with multimorbidity also preferred not to delegate decisions regarding their own EOL care unless they felt incapacitated [2,40], and expressed concerns about becoming a burden to others. Thus, our finding is not so surprising considering that increased comorbidity burden renders patients more likely to have DNR directives.

5. Limitations

This study has several limitations. First, its cross-sectional design prevented us from drawing causal inferences. Future longitudinal research should examine how functional and psychosocial factors influence informal ACP discussions and DNR directives over time.

Second, the 40 % nonresponse rate may have introduced selection bias and led to an overestimation of informal ACP discussion prevalence, as non-responders tended to have better ADL function. Traditional Taiwanese cultural norms often discourage open discussions about death, which may have influenced participants' willingness to engage in ACP-related conversations [41]. However, similar studies in both Western and Eastern contexts have also reported high nonresponse rates [2,21]. Future research should explore reasons for nonresponse and verify ACP discussion experiences.

Third, we did not examine the role of economic status, educational level, or behavioral readiness (e.g., transtheoretical model stages) in informal ACP discussion engagement. Additionally, while family and caregiver involvement are known to shape ACP decisions [3,9,21,42], we did not specifically investigate their influence in the Taiwanese context. Future studies should address these interpersonal and socioeconomic factors.

Lastly, our measurement of informal ACP discussions relied on a single yes/no question, which did not capture discussion depth/frequency/content. This may have led to an overestimation of ACP engagement, as we did not distinguish between casual and substantive discussions. Future research should adopt mixed-method approaches to provide a more nuanced understanding of the ACP process from informal ACP discussions to a DNR decision.

6. Conclusions and Implications

Understanding the prevalence of and factors associated with informal ACP discussion experience and DNR directives is crucial for designing effective interventions to increase formal ACP discussion and DNR directive uptake. According to our findings, geriatricians could tailor their strategies to promote formal ACP counseling by considering demographic, comorbidity, and functional factors. Patients with lower educational levels, preserved ADL function, and fewer comorbidities should be especially targeted. Additionally, fostering collaboration between healthcare systems and individuals can promote widespread access to respectful, well-informed EOL care for older adults with complex medical conditions. Future research should further investigate how cultural and family dynamics, as well as individual beliefs, shape ACP discussion and AD decisions.

Availability of data and materials

The data and resources were shared with eligible researchers through established academic channels. The datasets used in this study are available upon request from the corresponding authors.

Author contributions

Study concept and design: Yu-Tai Lo, Mei-Hua Chen, and Sheng-Yu Fan. Acquisition of data: Yu-Tai Lo and Mei-Hua Chen. Analysis and interpretation of data: Sheng-Yu Fan and Chung-Yi Li. Drafting of the manuscript: Yu-Tai Lo. Critical revision of the manuscript for important intellectual content: Deng-Chi Yang, Chi-Chang Huang, Chung-Yi Li, Mei-Hua Chen, and Sheng-Yu Fan.

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Conflicts of interest

This study was conducted in the absence of commercial or financial relationships that could be construed as potential conflicts of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jfma.2025.03.024>.

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